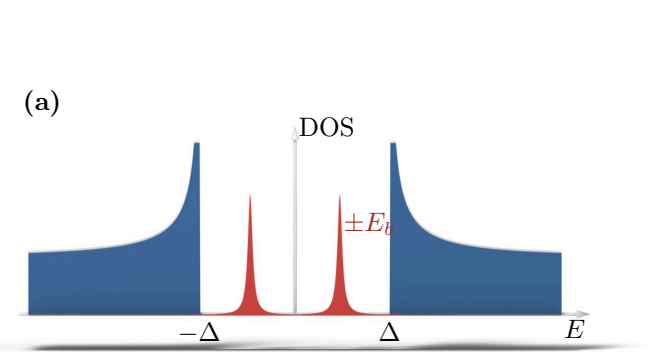
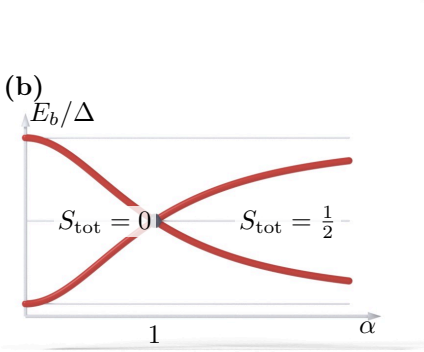

(a)
DOS
$\pm E_b$
$-\Delta$
Δ
E


(b)
E_b/Δ
$S_{\rm tot}=0$
$S_{\rm tot}=\frac{1}{2}$
1
α